A +5V_LOGIC (adapter ≥5 V/3 A) → U1 VIN, U2 5V, U7 VCC. Never feed it from BT1 or U10.
B +3V3 (U1 LDO) → U4, U5, U6, U8, R1/R2. HX711 and IR OUT stay ≤3.3 V (GPIO not 5 V tolerant).
C +VSERVO = BT1 (2S LiPo) → F1 → SW2 → U10 buck (5.5–6.0 V) → U9 only.
MG996R window 4.8–7.2 V: 1S LiPo too low, 2S full (8.4 V) too high, so buck required.
D GND_LOGIC and GND_PWR meet at ONE star point (NT1). Servo return current never crosses logic GND.
E UART2 (U1) ↔ UART1 (U2): TX→RX, RX→TX, shared GND. Both 3.3 V logic, no level shifter.
F Servo wires: twisted V+/GND pair ≤30 cm, away from HX711 / load-cell cable (cross at 90°).
G U2 needs 5 V ≥2 A with a local 470 µF cap; U2 IO0 floats in run mode (flash jumper removed).

ARCHITECTURE V2.1
U1 WROOM: RTC, OLED (SPI), HX711, servo, IR, HC-SR04, state machine
U2 CAM: Wi-Fi, web UI, camera, TinyML
U5 OLED: 7-pin SPI
Link: UART 115200 8N1

U4 DS3231 RTC
BT2 CR2032
VBAT (14)
VCC (2) SDA (15) SCL (16) SQW (3) 32K (1) GND (13)
+3V3
R2 4.7k
R1 4.7k
ZS-042 module: remove charge diode (CR2032)

U5 OLED SPI 7-Pin
VCC (2) GND (1) DIN (4) CLK (3) DC (5) RES (6) CS (7)

U6 XFW-HX711
LC1: Load Cell — RED E+ BLK E- WHT A+ GRN A-
Bowl rests on LC1 only (no frame contact)
VCC DT SCK GND

U7 HC-SR04 *hopper level*
VCC (1) TRIG (2) ECHO (3) GND (4)
R4 1k
R5 2k

U8 IR Obstacle *SSR1017*
VCC OUT GND

U1 ESP32-WROOM-32 Sensor Hub
3V3 (2) GND (1) GND (15) GND (38)
IO21 SDA (33) IO22 SCL (36) IO23 DIN (37) IO19 CLK (31) IO4 DC (26) IO32 RES (8) IO13 CS (16)
IO25 HX_DT (10) IO26 HX_SCK (11) IO27 TRIG (12) IO33 IR_OUT (9) IO34 ECHO (6)
SENSOR_VP SENSOR_VN VIN 5V IO35 IO17 TX2 (28) IO16 RX2 (27) IO18 SERVO (30) IO14 IO12 SD2 SD3 CMD CLK SD0 SD1 IO15 IO2 IO0 IO5 RXD0 TXD0
UART0 (USB) = debug/flash only
Strapping IO0/2/5/12/15 unused

U2 ESP32-CAM Vision/IoT Hub
5V GND
U0T GPIO1 U0R GPIO3 IO13 RX1 IO14 TX1 IO0 IO16 IO15 IO12 IO2 IO4 3V3
+5V_LOGIC
R5 1k
R6 1k
J3 = FTDI header, flash only;
IO0→GND while uploading, then remove
IO16 = PSRAM CS (unused); IO13/14 = SD pins, no SD

UART BRIDGE
115200 8N1 · 3.3 V logic
TX→RX and RX→TX, common GND

J3 FTDI_PROG
TX RX IO0 GND

DOMAIN A · LOGIC 5 V (never from LiPo/servo buck)
J1 5V ≥3A — +5V GND
C5 1000µF
C6 470µF (at U2 5V)
+5V_LOGIC
GND_LOGIC
NT1 Net-Tie_2
NT1 = ONLY tie between grounds
star point

DOMAIN C · SERVO (isolated)
BT1 2S LiPo 7.4V (+ / −)
F1 3A
SW2
U10 Buck 5.8V — IN+ IN− OUT+ OUT− — 5.5–6.0V ≥3A LDO
+VSERVO
GND_PWR
GND_PWR rail
servo return current stays on this rail
C10 2200µF
C11 0.1µF
SERVO PWM
U9 MG996R gate servo
SIG (OR) V+ (RD) GND (BR)

WIRE LEGEND
- signal / +3V3 / +5V
- UART bridge U1 ↔ U2
- servo domain (+VSERVO / GND_PWR)

Domain B: +3V3 from U1 LDO → U4, U5, U6, U8, R1, R2

Title:	**Smart Pet Feeder V2 - Dual MCU**
File: TL_CIRCUIT_V2_0.kicad_sch	Sheet: 1/1
Company: Tinkering Lab TL'26 · Size A3	
Date: 2026-10-14	**Rev: V2.1**